Project Vision - EDA on Cancer Dataset

This project harnesses the power of advanced analytics and global health data to uncover actionable insights into cancer care,

outcomes, and disparities. Using a robust dataset of 50,000 cancer patient records collected from multiple countries between

2015 and 2024, we aim to bridge the gap between raw data and life-saving decisions.

What the Data Covers

· Our dataset offers a 360-degree view of cancer patient profiles, including:

. Demographics: Age, gender, country, year of diagnosis.

. Genetic & Lifestyle Risks: Genetic predisposition, smoking, alcohol use, obesity

. Environmental Exposure: Air pollution

. Clinical & Economic Variables: Cancer type, stage, treatment cost

. Patient Outcomes: Survival years, severity scores

Core Objectives

1. Exploratory Data Analysis (EDA)

. Identify key trends, hidden patterns, and relationships

. Visualize disparities in diagnosis, lifestyle, treatment, and outcomes

. Highlight variations across countries, age groups, and cancer stages

2. Inferential & Predictive Analytics

Using statistical methods, we'll explore and answer critical healthcare questions:

. Determine the relationship between risk factors and cancer severity

. Analyze the proportion of early-stage diagnoses by cancer type

. Identify key predictors of cancer severity and survival years

. Explore the economic burden of cancer treatment across different demographics and countries

. Assess whether higher treatment cost is associated with longer survival

. Evaluate if higher cancer stages lead to greater treatment costs and reduced survival years

. Examine whether higher genetic risk amplifies the negative effects of smoking on cancer severity and survival outcomes